Student Handout

Name: _____ **Date:** _____ **Class:** _____

\\ MyDesign Portfolio

Element D Initial Template

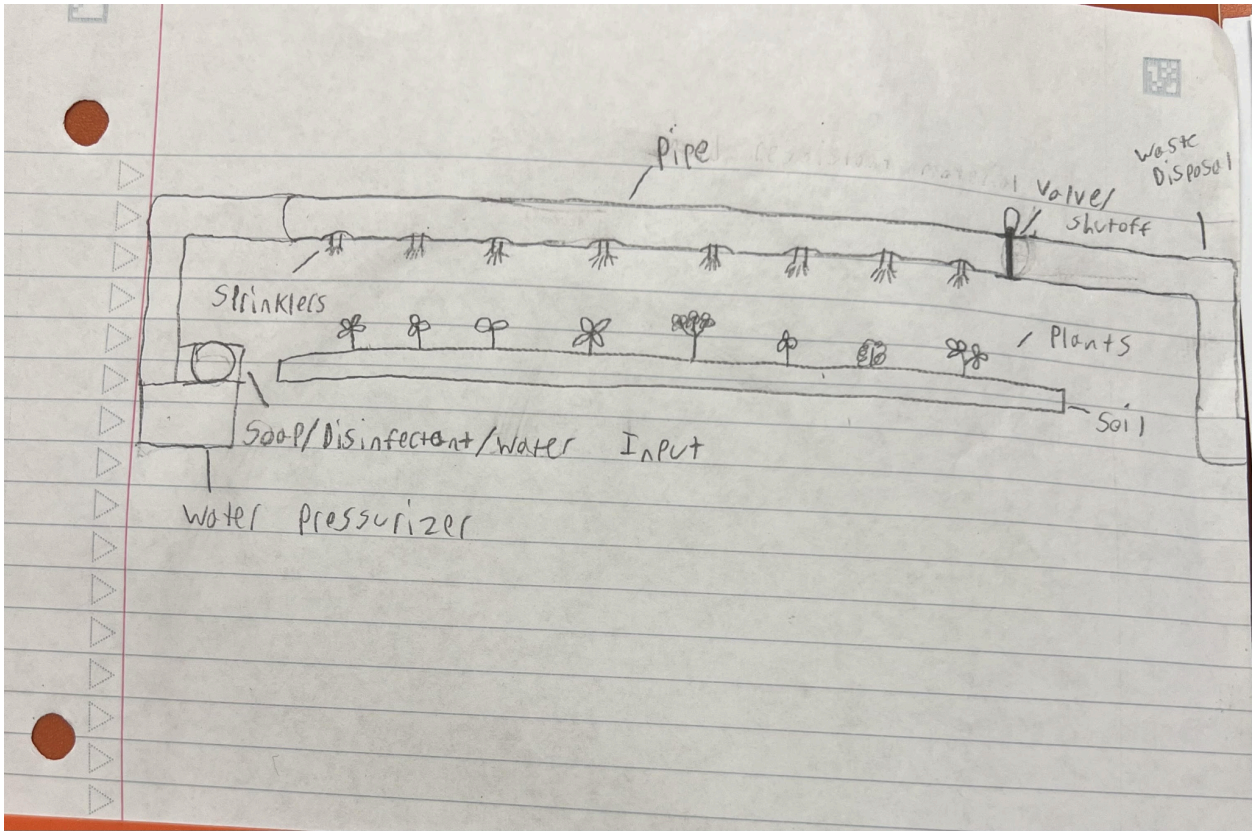
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1

Date: 3/10/25
Description: Disinfection system
Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____ **Date:** _____ **Class:** _____

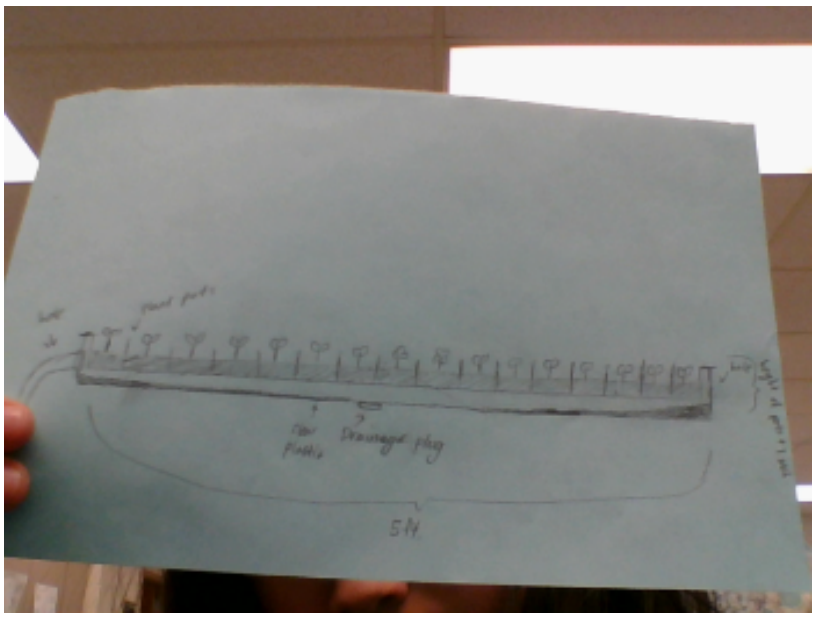


Design Idea #2

Date: 3/10
Description: A big tray that is double layered, the top layer has holes in the bottom so water can get through and will hold all the plant pots or just loose soil and plants. The outer wall will have a location for the hose to attach on one side and an opening on the other for excess water to flow out. The lower layer will have a slanted floor to direct the water to a second drainage hole at the bottom and will be clear(?). The top layer will also be detachable for easy cleaning.

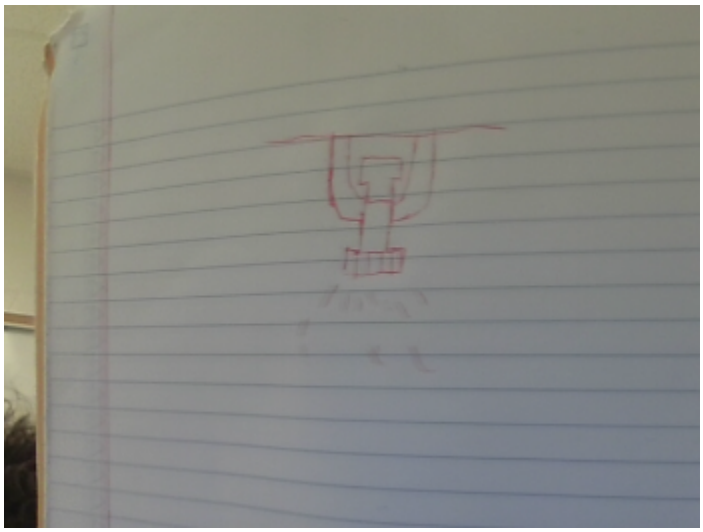
Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____ **Date:** _____ **Class:** _____



Design Idea #3

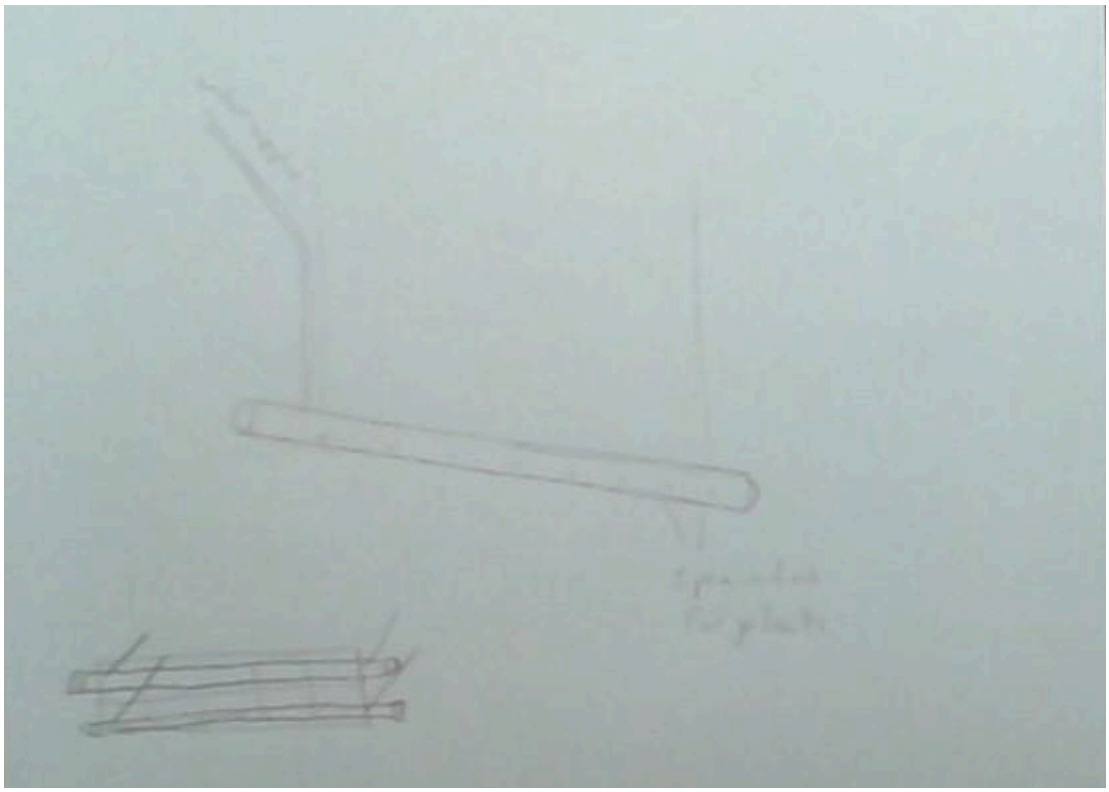
Date:3/10
Description:it is like a fire sprinkle but it does not go off for a fire it goes off when the plants need water.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Name: _____ **Date:** _____ **Class:** _____

Design Idea #4

Date: 3/10
Description: Angled PVC pipe helps provide water to each plant while preventing stagnant water buildup
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection *(D1, D2)*

	weight	Option 1	Option 2	Option 3	Option 4

Name: _____ **Date:** _____ **Class:** _____

Mold-less	5/5	1	1	1	1
Durable	4/5	1	1	1	1
Easy to use	4/5	-1	0	0	0
Even distribution	3/5	0	0	0	-1
Low cost	1/5	-1	-1	0	1
score		0	1	2	2

Our team chooses design 4 to solve the community problem because it fills the requirements of being moldless and durable while being cost-friendly and easy to make and replace.

Design Blueprint *(D4)*

Design Blueprint #1

Date: 3/12/25
Sketch or CAD Drawing:

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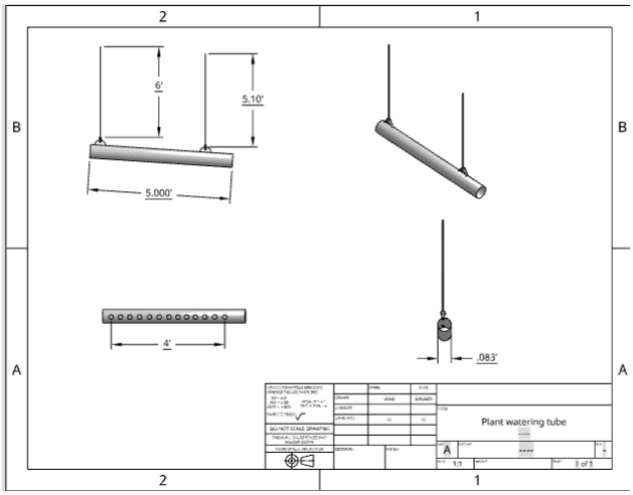

Manufacturing Methods:

Step 1: Purchase materials

Step 2: Drill holes into the PVC/ Calc Measurements

Step 3: Attach PVC pipe to wires and hang from ceilings

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
PVC pipe	free	1 or 2	0.00
Steel Wires(2 pc)	5.49$	1	5.49$
Drills	free	1	0.00
Total Cost	5.49	3/4	5.49$ before tax

Plan of Action *(D3)*

- Build a mini model and test for criteria(mold-less, durability, even distribution)
    - Or make several mini models to test with max efficiency
- Make changes based on information gathered
- Repeat until we run out of resources(time, supplies, patience) or we get a model that works good enough

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Iterations for the Solution:

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was *comprehensive, iterative, and consistently* defensible (viable design highly likely)	was *thorough, iterative, and* generally defensible (viable design likely)	was *adequate, generally* iterative and defensible *(viable design is possible)*	was *partial or overly general and only somewhat* iterative and/or defensible (raising issues with the design viability)	*was incomplete or minimally iterative and/or defensible (raising issues with the design viability)*	*was not evident that there was an attempt*
2.Design Solution Choice. The Design Solution chosen _____.	was *well-justified* and demonstrated attention to *all* design requirements	*was justified* and demonstrated attention to *most if not all* design requirements	*was explained with reference to at least some* design requirements	*was not sufficiently explained with reference to* design requirements	*lacked support related to* design requirements	*was not attempted or evidence not provided*
3. Plan of Action. The plan of action _____.	*has considerable merit and would easily*	*would* support repetition and testing	*might not clearly or fully support repetition*	*has insufficient detail to allow for testing for replication of*	*is lacking to the point that the design*	*cannot be executed or does not exist*

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	support repetition and testing for effectiveness by others	for effectiveness by others	*and testing for effectiveness by others*	*results*	*solution cannot be tested*	